Ice has latent heat of fusion . Meltwater at is not reused.

At pressure and volume , air has internal energy .

Good heat exchange keeps the air in the vessel isothermal during ejection.

Treat ejection as steps, each expelling volume , with far below the vessel volume. In each step, nozzle gas accelerates under the pressure difference and gains in kinetic energy and exits. Ignore its volume change while accelerating. Assume vessel pressure stays uniform during ejection; the outside pressure is atmospheric.

The electrical energy obtainable is of the sum of .

If , then .

Using these assumptions, find the electrical energy obtainable from the iceberg.

VI. (points) Large vertical parallel metal plates , are apart, at constant voltage . A metal-coated table-tennis ball of mass hangs between them. Nudge it towards one plate. It rebounds, hits the other plate, and repeats. Collisions are inelastic, with restitution coefficient . Contact is very brief, during which electrostatic equilibrium is reached. The ball's charge relates to plate voltage by , where is constant. Assume the ball radius is far below plate spacing , and its charge does not affect plate charge distribution. A long enough string makes the motion nearly horizontal and straight. Neglect initial kinetic energy at the first collision and air resistance. For the setup shown, find: